… Lecture 5 … Shrinkage

Part 1: uncovering the truth behind random effects!

Fixed or Random

- What is a fixed effect?
- What is a random effect?
- Why does it matter?
- What is the difference?

In setting out a model, decisions need to be taken on whether a factor is fitted as a fixed effect or a random effect. Does it matter? Of course it does!

We can see one difference in what emerges from the analyses with fixed and random factors: whereas both give us estimates of effects, when a factor is declared as random, then an estimate of a variance is also produced and, if more than one trait, covariances as well.

Random Effects

- Randomly drawn from a distribution with a finite variance
  - » typically Multi-Variate Normal … MVN
  - » but doesn't have to be!
  - » a form of prior knowledge that can be used
  - » e.g. in BLUP, REML

First, recap what is a random effect? My definition is that if we can assume the effects are drawn at random from a distribution with an assumed form of (co)variance matrix, then it can be fitted as random. The distributions are typically assumed to be Multivariate Normal (MVN). Multivariate, because we have different levels of the random factor, and the value for each level is assumed to be a draw from the random distribution, so we have multiple draws from the distribution and different levels may have non-zero covariances between them, so not independent – we have seen that with **A**. It does not have to be Normal and we will come across models that do not assume this later in the course.

So when a term is put in a model as a random effects we are making an assumption that the effects are drawn at random from a distribution with a finite variance, typically MVN.

My definition of fixed effects is 'not random'!

Fixed Effects

- Randomly drawn form a distribution with a finite variance
  - » typically MVN
  - » but doesn't have to be!
  - » a form of prior knowledge that can be used
  - » e.g. BLUP, REML
- Fixed effects
  - » not Random!

Examples of Fixed Effects

- Systematic, repeatable across experiments
  - » e.g. sex, drug, birth type
  - » effects of temperature, or light
  - » top two sires and bottom two sires
  - » "almost all levels of factor contained in the data"

Some examples of effects that are always fixed are systematic and repeatable effects such as differences among sexes, age groups, birth types (singles and twins), drug doses, fertiliser treatments, covariates such as the effect of light or temperature etc.

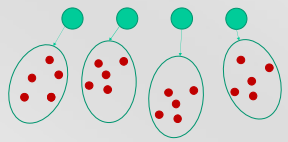
We have treated sires as a classical random effect, but it could be fixed sometimes! For example, suppose we rank the sires and take the top and bottom sires in a distribution then they cannot be treated as randomly sampled!

The definition of fixed as "almost all levels are in the data" is a start, but does not really go far enough.

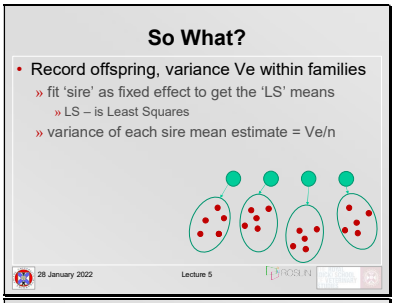
So what difference does this make? I am going to give a 'heuristic' argument, which is an argument that is not formally correct but illustrates the principles correctly.

So What?

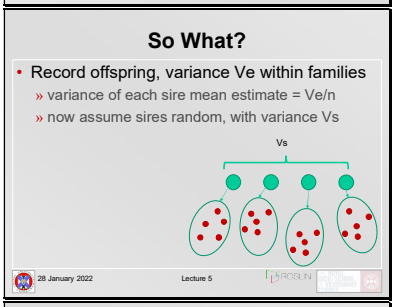
- Imagines some sire families
  - » each with n offspring … e.g. n=5 here



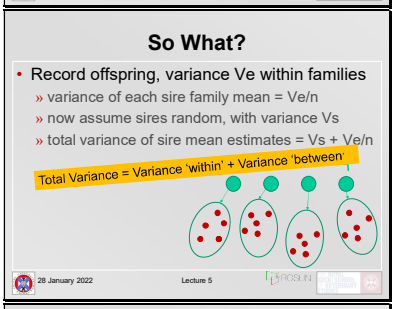
Imagine some *randomly* sampled sires, and each of them have *n* offspring. Suppose the offspring are measured and the simple means of the sire families are calculated.



These means are what we would get by fitting the sires as 'fixed' and so getting the 'Least Squares Means'. Each of the means will have a sampling variance of *Ve/n* where *Ve* is the variance within sire families.

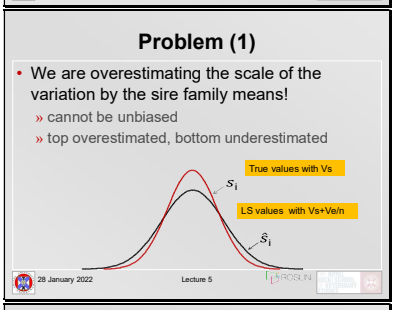


If we assume sires are random, we assume their true means have a finite variance, *Vs*.

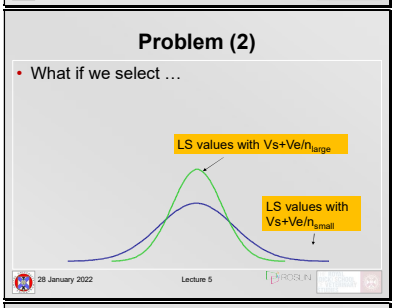


The total variance is the 'variance within groups' plus the 'variance between groups'.

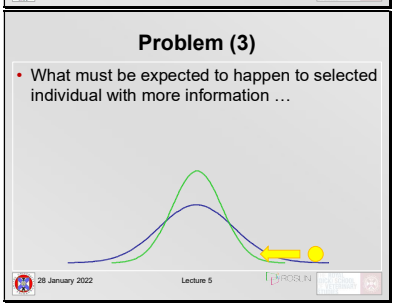
Therefore for our total variance of the estimated means is then *Vs*+*Ve/n* which is obviously > *Vs*!



So we can immediately see a problem as our estimates have more variance than what we expect. Consequently, we must be overestimating and underestimating the extremes.

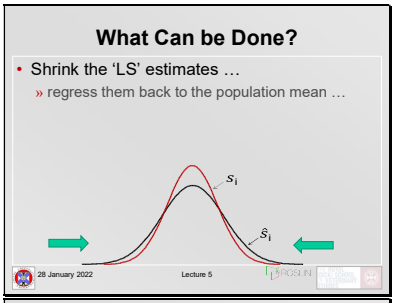


If we had a mix of different numbers of offspring, then the extremes will be dominated by those that have fewer offspring as they have more sampling variance, not necessarily are the best. So the individuals with least information will tend to be selected, or identified for whatever purpose as being extreme.

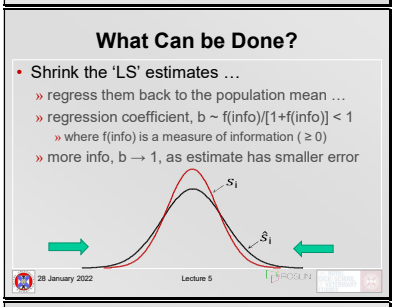


What is then expected to happen, as more information is obtained, is that the 'best' individual moves back towards the mean – as its superiority came from positive errors, and not from its intrinsic superiority.

In passing, I note that this was a serious problem in the early days of genetic prediction in livestock, when models used fixed effects. The best dairy bulls would be identified and promoted to breeders, who then followed the advice and used them for breeding stock. As illustrated above these bulls would tend to be those with the least information. More often than not, at the next evaluation with more information, the breeders found that the bull would be revealed to be much closer to average. This brought the idea of genetic prediction into disrepute – thankfully rescued by more enlightened (mixed) models.

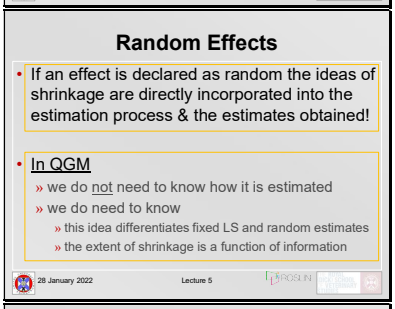


To solve this problem we can **shrink** what we get from estimating the simple (Least Squares) means. We 'regress them back to the 'population mean' using a regression coefficient *b* that is < 1.

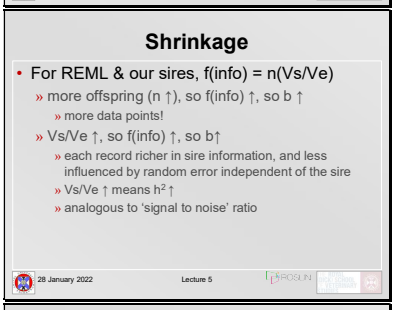


We shrink them by an amount that depends on the amount of information we have available. The more information we have, the smaller the sampling error, and the more certain we are of that the simple mean represents the true merit, so we are less willing to shrink them very far, so *b* is closer to 1.

The form of this function for assuming the effects are MVN is $f(info)/(1+f(info))$ which tends to 1 as the information accumulates.

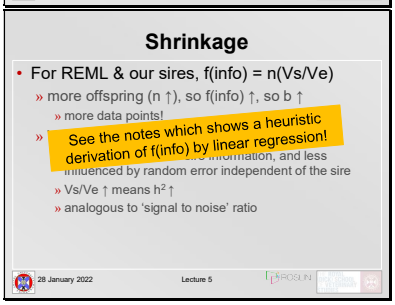


The key points to remember for QGM are shown here:
1. If an effect is declared as random then the ideas of shrinkage are directly incorporated into the estimation process.
2. We do not need to know the details of how these estimates are made.
3. We do need to know (i) the idea of shrinkage differentiates the least squares (LS) estimates used for fixed effects and the estimation of random effects, and (ii) the extent of shrinkage is a function of information on the effect.



Therefore the effect of saying we have random effects is that what would have been estimated as a fixed effect is regressed back to the population mean. The more information the regression is less.

In the case of the sire families, more information is (i) having more offspring or (ii) having a greater *Vs/Ve* ratio. The latter is equivalent to a higher heritability, or a greater signal to noise ratio i.e. each offspring record contains more information on the sire.



It is highly recommended that students explore this idea in more detail by reading 'shrinkage_heuristic.pdf' which extends the sire family analogy. 'Heuristic' means that it comes to the right answer using sound principles, but is 'hand-waving' rather than statistically rigorous. As mentioned above, the estimation is seamless rather than the idea of a two steps in the heuristic argument.

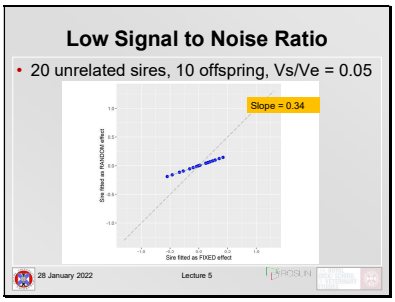
One question that has been asked in the past is how it relates to fitting an Individual Model, so 'shrinkage_heuristic_Amatrix' explores shrinkage in the context of BLUP – it is the same idea but takes a bit more effort to show, but worth looking at.

To illustrate these ideas further the following plots of effects estimated when sires are treated as random (y) plotted against the effect estimated when sires are treated as fixed (x), using data simulated in R.

Firstly assume 20 unrelated sire, each with 5 offspring, and Vs/Ve = 0.05, which is equivalent to saying $h^2 \approx 0.19$.

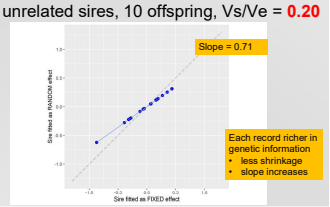
The regression back is shown … in this example, the slope is 0.34 < 1, i.e. 'SHRINKING' effects.

(Note: the plot show a nice straight line because all individuals have the same information, and so are shrunk by the same amount).



Increased Signal …

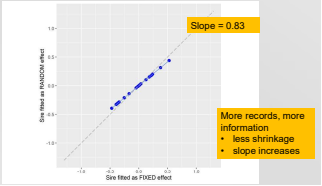
- 20 unrelated sires, 10 offspring, Vs/Ve = **0.20**



Now if the information is increased by making each offspring record richer in genetic information by making Vs/Ve = 0.2, or $h^2 \approx 0.67$, the slope increases to 0.71 – more information, less shrinkage, closer alignment to the LS estimate.

More Records …

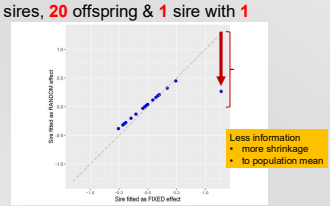
- 20 unrelated sires, **20** offspring, Vs/Ve = 0.20



Now if the information is increased by more records, by having 20 offspring per sire, the slope increases to 0.83.

Applied to Each Effect …

- **19** sires, **20** offspring & **1** sire with **1**

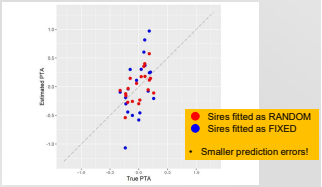


Remember shrinkage is applied to each effect based on the information available on that effect alone, irrespective of how much information is available on other effects!

Here one of the 20 sires now only has one offspring (it happened to have the best LS estimate!) and the others still have 20. It is clear that the shrinkage is much stronger on the sire with one offspring than on the others.

Benefits

- Comparison of prediction errors



The point of this plot is just to note that the prediction errors are smaller when fitting sires as random compared to the LS fixed estimates. So the random model is doing a better job.

Summary

- Random instead of fixed leads to <u>shrinkage</u>
  - » regression back to the distribution mean …
  - » regression coefficient, b ~ f(info)/[1+f(info)] < 1
  - » more info, f(info) increases and b → 1
    - » the estimate has smaller random error
  - » b → 1 with more data and greater 'signal to noise ratio' per data point
  - » shrinkage is differentially applied effect by effect
  - » see 'revision_5_heuristic.pdf'
  - » see 'revision_5_heuristic_Amatrix.pdf'

So when we assume that effects are drawn at random from some distribution with a finite variance then it results in 'shrinkage', where estimates are regressed back to the population mean, more strongly as the available information in the effect decreases. With no information the estimate of the random effect is 0, and the fitted value is determined by the other fixed and random effects in the model.

… Lecture 5 …
Shrinkage

Part 2: the origins of REML

REML was introduced by Patterson and Thompson (at the University of Edinburgh!) in Biometrika (1971) in a paper entitled "Recovery of inter-block information when block sizes are unequal". This issue arises when the 'blocks' in a trial (blocks are 'nuisance' factors) are unbalanced and hold different numbers of each treatment. If block '1' has an excess of treatment 'A' say, and another block '2' has an excess of treatment 'B', what weight should be given to the contrast of the mean of block '1' with the mean of block '2' for providing information on treatments 'A' and 'B'?

Model Terms

- Models will have different types of terms
  - » factors of interest e.g. genetics
    - » concerned with hypotheses being tested
      - » fixed effect contrasts of breeds
      - » random effect estimates of genetic variance
  - » nuisance factors
    - » would not be present in an ideal experiment
    - » e.g. 'blocks', fields, hospitals, herds, tanks, batches
  - » a genetic model is only as good as its associated environmental model!

When building a model for analysis there will be a variety of factors and covariates to account for. When thinking about these terms it is important to consider what in the model is a terms of interest for our hypotheses (i.e. typically genetic for geneticists!) and what are 'nuisance' factors. The latter are terms that would not exist in an ideal trial, but for one reason or another exist and have to be dealt with – remember a genetic model is only as good as how well it accounts for the environment i.e. the environmental model. The variation arising from nuisance factors that are not, or inadequately, accounted for may erroneously influence (i.e. bias) the terms of interest, or reduce the power to detect them.

Modern genetic datasets, e.g. medical data, are often huge and one should not be miserly giving degrees of freedom in modelling nuisance parameters to remove possibilities of bias in the terms of interest. When datasets become small, this becomes more challenging.

Fixed or Random

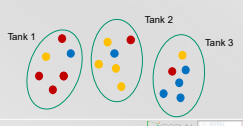
- Choice of fixed/random depends on context
  - » not automatic although often 'routine'

When it comes to deciding whether a factor is fixed or random, it may seem initially that somehow this is automatic. It is not! While it is often routine, it will always depend on context and objectives.

One point to remember is that if a term is changed from fixed to random, not only is an assumption being made on randomness and a finite variance, but also shrinkage will then be applied. Thinking of this as regressing a LS estimate back to the population mean, then the shrinkage will release some variance (that would have been explained by the effect when using LS) to be interpreted by other effects in the model. In our simple model consisting only of sires and residual this 'released' variance ends up in the residual.

Recovery of Inter-block Information

- Patterson & Thompson's original REML paper was on recovery of inter-block information
  - » illustrates consequences of random/fixed choice
  - » incomplete block designs e.g. fish tanks
  - » e.g.

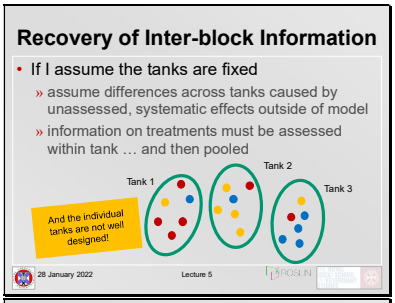


Consider an experiment with 3 treatments red blue and yellow, but with a nuisance factor e.g. 3 'tanks' in an aquatic trial.

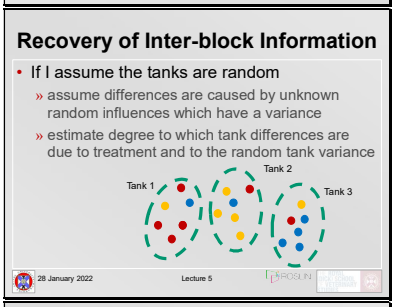
Recovery of Inter-block Information

- If I assume there is no difference among the tanks
  - » i.e. if variation among tanks is small
  - » … I have a good experiment!

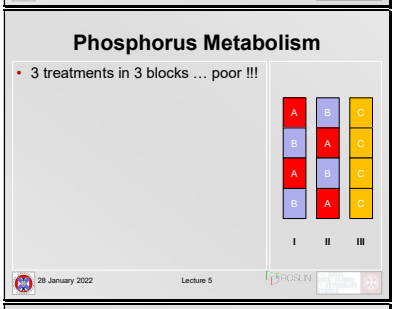
If there are no tanks, then the design is balanced and treatments effects are estimated very efficiently.



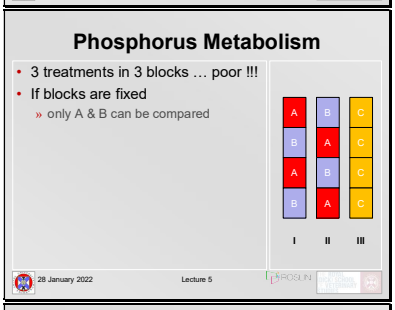
If I choose tanks as ‘fixed’ I cannot recover any information from between blocks, I can only look at the information available within each tank, and then ‘pool’ it across the tanks – e.g. I can get an estimate of (‘red’ – ‘yellow’) from within each tank, and take a weighted average of them; weighted because the estimates from different tanks will have a different sampling variance, some more precise than others. Here all these estimates are very poor – tank 1 has 4 red and 1 yellow, tank 2 has 4 yellow and 1 red, and tank 3 1 red and 1 yellow. The overall impact is to give a s.e. 10% larger than the ideal experiment.



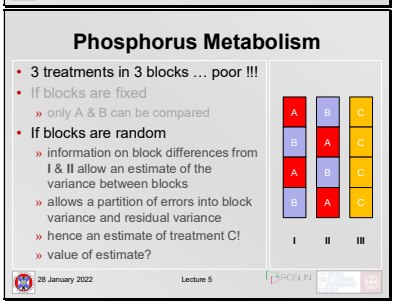
If I choose tanks as ‘random’, then the tanks effects are regressed back towards the mean, releasing some variation to be interpreted as potential treatment effects. Therefore I can use e.g. the difference between the means of Tank 1 (dominated by ‘red’) and Tank 2 (dominated by ‘yellow’) in estimating (‘red’-‘yellow’). The weight that this information is given compared to the estimates made within tanks will depend on the magnitude of the estimate of the tank variance.



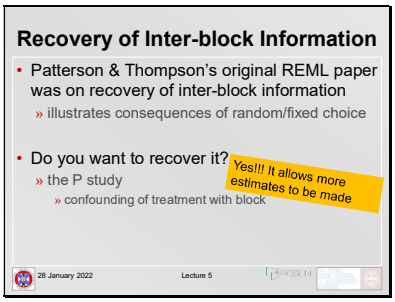
An illustration of this recovery is from a (real) dietary experiment on Phosphorus metabolism in sheep – the design is very poor, although there were reasons for it! There were three treatments ‘A’, ‘B’ and ‘C’ and the three blocks. The first two blocks have a good design for treatments ‘A’ and ‘B’, but ‘C’ and block ‘III’ are completely confounded.



If blocks are treated as fixed then only ‘A’ and ‘B’ can be compared. No statements can be made about ‘C’ in comparison to ‘A’ and ‘B’ – are any differences due to block or are they due to ‘C’?



If blocks are random, then the means for blocks **I** and **II** can be used to estimate a variance between blocks (albeit a poor one with 1 d.f.) and an estimate of ‘C’ compared to ‘A’ and ‘B’. If this variance is small and the difference between block **III** and the others is large, then there is evidence that treatment ‘C’ caused the difference observed; on the other hand if the variance between blocks **I** and **II** was large compared to the difference between **III** and the other blocks, then any evidence on the effect of ‘C’ is very weak. However with blocks as random estimates of ‘A’-‘C’ and ‘B’-‘C’ are available, and their s.e.s will depend on the estimate of block variance.

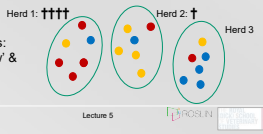


Do I wish to recover the inter-block information? Definitely YES! Some objective information is better than no information.

Genetic Variation in TB

- A bovine TB breakdown is a mini-epidemic
- Herds are closed during epidemic (H1, H2 …)
- How does one interpret the following?

Sire families: 'red', 'yellow' & 'blue'



Now consider another real example of genetic variation in bovine TB, a serious problem in UK cattle. A bovine TB outbreak can be considered as a mini TB epidemic as when a case is identified, the herd is closed and isolated from other cattle.

In the design, Herd 1 has a lot of TB mortality compared to Herds 2 and 3. Herd 1 has many more offspring of sire 'red' compared to the other herds. Can this be interpreted as 'red's offspring being more susceptible?

Genetic Variation in TB

- Are herds fixed or random?
  - » is it OK to say "H1 high TB rate and ●", "H3 with low TB rate and ●", gives evidence "● > ●"?
  - » i.e. is it OK to recover information from between herds?
  - » i.e is it OK to treat herds as random from a distribution?

Is it a good idea to declare herds as random and recover the 'inter-herd' information and use it to estimate genetic variation?

Genetic Variation in TB

- Are herds fixed or random?
  - » why has Herd 3 no cases?
  - » is it due to 'blue' sire or absence of pathogen?
  - » no badgers?
  - » "forces of infection", importation events
- Need an excellent between herd model!
  - » otherwise you interpret unknowns as genetics

Here the answer is NO! Is Herd 3 TB free because it has no pathogen, or due to the 'blue' sire, or the absence of badgers - or is Herd 1 suffering because it imported a cows from a heavily infected herd?

If herds are treated at random, then the between-herd variation will be used to estimate genetic effects – but how good is the between-herd model? There is a big risk of biasing estimates of genetic variation by incompleteness and inadequacies in modelling the effects that systematically differentiate between herds. A genetic model is only as good as its environmental model.

The spread of bovine TB is a complex issue, and the genetics of infectious diseases is surprisingly controversial and often neglected in epidemiology. A study using between-herd information for genetics of bovine TB would be open to severe criticism, whereas when herds are fixed only information within herds is used and then pooled across herds.

So fixed or random depends on context and objectives. Assuming a factor is random may be acceptable or desirable in one context but not in another.

Recovery of Inter-block Information

- Patterson & Thompson's original REML paper was on recovery of inter-block information
  - » illustrates consequences of random/fixed choice
- Do you want to recover it?
  - » the P study
    - » confounding of a treatment with block
  - » the TB study
    - » allowing infor another

Yes!!! It allows more estimates to be made

No! The assumptions make the findings more vulnerable to model error.

… Lecture 5 …
Shrinkage

Part 3: some generalities

Genetic Models

- Often genetic effects treated as random
- Sire models
  - » assume sire effects are $\sigma_s^2 \mathbf{I}$ with $\sigma_s^2 = ¼\, \sigma_A^2$
    - » σ_A^2 is the additive genetic variance
  - » or are $\sigma_s^2 \mathbf{A}$ (as Tutorial 1!)
    - » **A** is Numerator Relationship Matrix
- Individual Animal model
  - » assume genetic effects of individuals are $\sigma_A^2 \mathbf{A}$
  - » or with genomic data $\sigma_G^2 \mathbf{G}$

Depending on the source of information, genetic effects may be fixed or random. In sire and individual models the genetic effect is random.

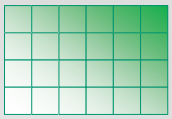
Genetic Models

- 'Genetics' is not always random
  - » may sometimes be fixed
  - » e.g. breed is typically fixed
  - » next lecture

Sometimes genetic information comes from breed. That is rarely fixed as the breeds in an experiment are rarely drawn at random from the breeds available.

Environmental Models

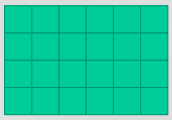
- Spatial models, e.g. pens in a shed
  - » or desks in a classroom
  - » or plots/quadrats in a field or forest
  - » or tanks in aquaculture shed



Spatial models occur in many kinds of study. Consider pens in a livestock shed.

Environmental Models

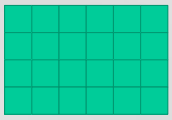
- fixed = … + pen
  - » here 23 d.f.



In the 6x4 set of pens, if pens are fixed then there are 23 d.f. for pens.

Environmental Models

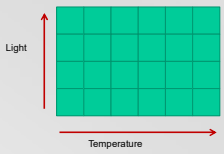
- fixed = … + row + column, random = … + pen
  - » here 8 d.f. (3 for row, 5 for column) & V_{pen}



One could try and uncover some of the structure of the pen effects e.g. one could fit 2 fixed effects 'row' (3 d.f.) and 'column' (5 d.f.), and have 'pen' as a random effect to estimate V_{pen}, the residual variation not explained by 'row' + 'column'

Environmental Models

- fixed = … + x_{light} + x_{temp}, random = … + pen
  - » here 2 d.f. & V_{pen}



Perhaps there are important environmental descriptors such as light and temperature measured for each pen. Then one could fit x_{light} and x_{temp} as covariates (2 d.f. in total) together with 'pen' as random to estimate the residual V_{pen}.

Or polynomials of x_{light} and x_{temp} and 'pen' …

Conclusion

- An effect can be described by a combination of fixed and random components!

The conclusion of the examples above is that an effect such as 'pen' can be described by appropriate combinations of fixed and random effects to uncover systematic effects in the variation observed!

Fixed & Random Interactions

	Factor 1	
Factor 2	Fixed	Random
Fixed	Either	Random
Random	Random	Random

- Two-way
  - » if one factor random, interactions must be random
  - » if factors are fixed then choice
- Multi-way
  - » if any lower order interaction is random, then high order is random

A technical point on interactions. If any factor or lower order interaction is random then the interaction must be fitted as random.

Summary

- Choice of fixed or random influences how information is used
  - » fixed or random is not absolute
    - » using a factor as random is making an assumption about the nature of the effect
    - » depends on the modelling consequences of making the assumption
- A genetic model is only as good as its associated environmental model!

The take-home summary!

A final thought for the more statistically minded. In the first part of the lecture it was shown that the regression coefficient for shrinkage was $b = f(info)/(1+f(info))$, with $f(info) = nVs/Ve$. If Vs/Ve becomes very large, then *b* approaches 1 irrespective of *n*, and an estimate always approaches the LS estimate. Therefore one description of a fixed effect is a random effect with an infinite variance. In many packages using MCMC, particularly with a Bayesian perspective, a fixed effect is fitted as a random effect with a very large variance. We take a brief look at MCMC and Bayesian statistics in Tutorial 10 using Jarod Hadfield's MCMCglmm package!